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```
package CM1;
```

```
public class Mahasiswa08 {
```

```
    String nim, nama, prodi;
```

```
    Mahasiswa08(String nim, String nama, String prodi) {
```

```
        this.nim = nim;
```

```
        this.nama = nama;
```

```
        this.prodi = prodi;
```

```
    }
```

```
    void tampil() {
```

```
        System.out.println("NIM : " + nim + " | " + "Nama : " + nama + " | " + "Prodi : "+  
prodi);
```

```
    }
```

```
}
```

```
package CM1;
```

```
public class Buku08 {
```

```
    String kode, judul;
```

```
    int tahun;
```

```
    Buku08(String koBu, String jdl, int thn) {
```

```
        this.kode = koBu;
```

```
        this.judul = jdl;
```

```
        this.tahun = thn;
    }
```

```
void tampil() {
    System.out.println("Kode : " + kode + " | " + "Judul : " + judul + " | " + "Tahun : " +
    tahun);
}
}
```

```
package CM1;
```

```
public class Peminjaman08 {
    String nim, nama, judul;
    int lamaPinjam;
    int denda;

    Peminjaman08(String nim, String nama, String jdl, int lm) {
        this.nim = nim;
        this.nama = nama;
        this.judul = jdl;
        this.lamaPinjam = lm;
        hitungDenda();
    }
}
```

```
void hitungDenda() {
    int batas = 5;
    int telat = lamaPinjam - batas;
    if (telat > 0) {
        denda = telat * 2000;
    } else {
        denda = 0;
    }
}
```

```

    }
}

void tampil() {
    System.out.println(nama + " | " + judul + " | " + "Lama : " + lamaPinjam + " Terlambat |
Denda: " + denda);
}
}

```

```
package CM1;
```

```
import java.util.Scanner;
```

```

public class MahasiswaMain08 {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Mahasiswa08[] mhs = {
            new Mahasiswa08("22001", "Andi", "Teknik Informatika"),
            new Mahasiswa08("22002", "Budi", "Teknik Informatika"),
            new Mahasiswa08("22003", "Citra", "Sistem Informasi Bisnis")
        };

        Buku08[] buku = {
            new Buku08("B001", "Algoritma", 2020),
            new Buku08("B002", "Basis Data", 2019),
            new Buku08("B003", "Pemrograman", 2021),
            new Buku08("B004", "Fisika", 2024)
        };
    }
}

```

```
Peminjaman08[] pinjam = {  
    new Peminjaman08("22001", "Andi", "Algoritma", 7),  
    new Peminjaman08("22002", "Budi", "Basis Data", 3),  
    new Peminjaman08("22003", "Citra", "Pemrograman", 10),  
    new Peminjaman08("22003", "Citra", "Fisika", 6),  
    new Peminjaman08("22001", "Andi", "Basis Data", 4)  
};
```

```
int pilih;
```

```
do {  
    System.out.println("\n=== MENU ===");  
    System.out.println("1. Tampilkan Mahasiswa");  
    System.out.println("2. Tampilkan Buku");  
    System.out.println("3. Tampilkan Peminjaman");  
    System.out.println("4. Urutkan Berdasarkan Denda");  
    System.out.println("5. Cari Berdasarkan NIM");  
    System.out.println("0. Keluar");  
    System.out.print("Pilih: ");  
    pilih = sc.nextInt();
```

```
switch (pilih) {
```

```
    case 1:
```

```
        System.out.println("\nData Mahasiswa :");  
        for (Mahasiswa08 m : mhs) m.tampil();  
        break;
```

```
    case 2:
```

```
        System.out.println("\nData Buku :");  
        for (Buku08 b : buku) b.tampil();
```

```
break;
```

case 3:

```
System.out.println("\nData Peminjaman :");  
for (Peminjaman08 p : pinjam) p.tampil();  
break;
```

case 4:

```
System.out.println("\nDenda :");  
for (Peminjaman08 p : pinjam) p.tampil();  
  
for (int i = 1; i < pinjam.length; i++) {  
    Peminjaman08 temp = pinjam[i];  
    int j = i - 1;  
    while (j >= 0 && pinjam[j].denda < temp.denda) {  
        pinjam[j + 1] = pinjam[j];  
        j--;  
    }  
    pinjam[j + 1] = temp;  
}
```

```
System.out.println("\nSetelah Sorting :");  
for (Peminjaman08 p : pinjam) p.tampil();  
break;
```

case 5:

```
for (int i = 1; i < pinjam.length; i++) {  
    Peminjaman08 temp = pinjam[i];  
    int j = i - 1;  
    while (j >= 0 && pinjam[j].nim.compareTo(temp.nim) > 0) {
```

```
        pinjam[j + 1] = pinjam[j];  
        j--;  
    }  
    pinjam[j + 1] = temp;  
}
```

```
System.out.print("Masukkan NIM: ");  
String cari = sc.next();
```

```
int left = 0, right = pinjam.length - 1;  
boolean ketemu = false;
```

```
while (left <= right) {  
    int mid = (left + right) / 2;  
  
    if (pinjam[mid].nim.equals(cari)) {  
        pinjam[mid].tampil();  
        ketemu = true;  
        break;  
    } else if (pinjam[mid].nim.compareTo(cari) > 0) {  
        right = mid - 1;  
    } else {  
        left = mid + 1;  
    }  
}
```

```
if (!ketemu) {  
    System.out.println("Data tidak ditemukan");  
}  
break;
```

```
}
```

```
} while (pilih != 0);
```

```
sc.close();
```

```
}
```

```
}
```